Abstract

on

PREDICTIVE GRID INTELLIGENCE FRAMEWORK FOR DISTRIBUTION SUBSTATION MAINTENANCE FAILURE PREVENTION THROUGH ADVANCED MACHINE LEARNING ANALYTICS

An abstract submitted for

Project Phase-I

By

Mr.Mohammed Abdul Sultan Akmal (160923737104)

Mr. Mohammed Abdul Rahman (160923737111)

Batch Number: B17

B.E.-I.T.-IV Year

Under the Guidance of

Mr.M Shiva Rama Krishna
Assistant Professor



DEPARTMENT OF INFORMATION TECHNOLOGY

LORDS INSTITUTE OF ENGINEERING AND TECHNOLOGY

(UGC Autonomous)

Approved by AICTE | Affiliated to Osmania University | Estd.2003 | Accredited by NBA & 'A' grade by NAAC

SY.NO.32, Himayathsagar, Golconda Post, Near Appa Junction, Hyderabad-500091, India.

A.Y. 2025-26

Project Title : PREDICTIVE GRID INTELLIGENCE FRAMEWORK FOR DISTRIBUTION SUBSTATION MAINTENANCE FAILURE PREVENTION THROUGH ADVANCED MACHINE LEARNING ANALYTICS

Batch Number: B17

ABSTRACT

Artificial Intelligence and Machine Learning have become essential technologies for predictive maintenance and fault diagnosis in power systems. Electrical transformers are critical assets in power distribution networks, and unexpected failures can result in significant operational losses and maintenance costs. Early fault detection and estimation of equipment health are therefore important for improving reliability and reducing downtime.This project proposes an intelligent Transformer Fault Detection and Remaining Useful Life (RUL) Prediction System developed using Django and Machine Learning techniques. The system analyzes Dissolved Gas Analysis (DGA) data collected from transformers and applies advanced machine learning models including LightGBM and XGBoost for fault classification and health assessment.The platform automatically identifies transformer fault conditions such as Normal Operation, Partial Discharge, Thermal Fault, and Electrical Fault. In addition to fault diagnosis, the system predicts the Remaining Useful Life (RUL) of transformers and calculates a Health Score that indicates the current operational condition of the equipment.To improve transparency and interpretability, Explainable Artificial Intelligence (XAI) techniques using SHAP analysis are integrated to identify the most influential factors affecting predictions. Interactive dashboards provide visualization of dataset analysis, model performance, fault probabilities, health status, gas trends, and prediction reports.The system supports user authentication, dataset management, real-time prediction, historical analysis, and administrative monitoring through a web-based Django platform. The proposed solution provides an efficient, scalable, and intelligent framework for transformer condition monitoring, predictive maintenance, and reliability enhancement in power distribution systems.

Keywords : Smart Grid, Distribution Substation, Predictive Maintenance, Machine Learning, Failure Prediction, Grid Intelligence, Fault Detection, Data Analytics, Equipment Monitoring, Failure Prevention

Mr. Mohammed Abdul Sultan Akmal (160923737104)

Mr.Mohammed Abdul Rahman (160923737111) **Mr. M Shiva Rama Krishna**